SPARKY'S SCIENCE FACTS

Why Are Some Sounds High and Some Low?

Sound · Ages 7+

> **Short answer:** A sound is high when something vibrates very fast, and low when it vibrates slowly. A sound is loud when the vibration is big.

How it works

- How high or low a sound is, is called its pitch.
- Short, thin or tight things vibrate fast and make high sounds. Long, thick or loose things make low sounds.
- How loud a sound is depends on how big the vibration is. A hard pluck makes a big, loud vibration.
- People can hear many pitches, but dogs can hear sounds too high for us.

Key words: Pitch: how high or low a sound is; Volume: how loud or soft a sound is

Common mix-ups

- Loud does not mean high: a big drum can be very loud and very low.
- Tapping a bottle with more water gives a lower note, but blowing across it gives a higher note.

Did you know? On a marimba, the long wooden bars play the low notes and the short bars play the high notes.

Try it: Fill glass bottles with different amounts of water and tap them gently with a spoon. Can you play a tune? Do it with a grown-up, as glass can break.

Ask Sparky: Why does a guitar have thick and thin strings? What is an echo?